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References:

1. Ajoy Ghatak, "Optics", Publisher Tata Mc Grew Hill, fifth Edition
2. A. Ghatak, K. Thyagarajan, Introduction to Optical Fiber Optics, Publisher, Cambridge University Press.
3. A. Beiser, "Concept of Modern Physics", Publisher, Tata Mc Graw Hill, Fifth Edition.

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21/3/2023
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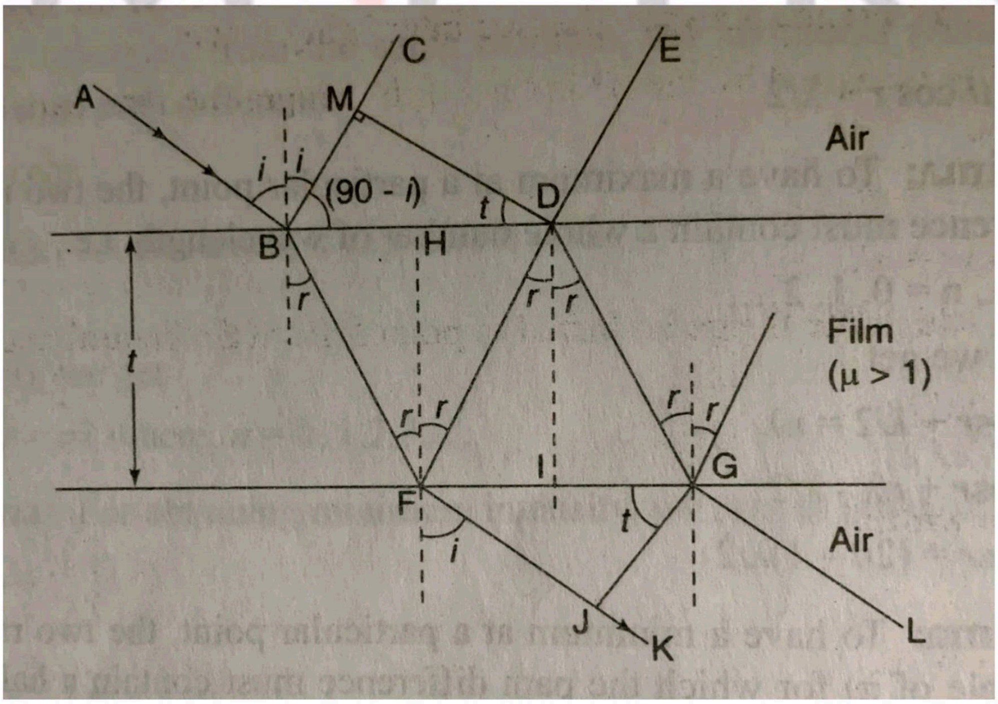
LECTURE -1

Concept of Interference

When two or more waves interact with each other with the principle of superposition in order to form new wave pattern is called interference.
Whenever crest or trough of one wave superimposes on crest and trough of another wave, wave of maximum intensity is formed, and the respective fringe will be bright. (Constructive interference)
Whenever the crest or trough of one wave interacts with trough or crest of another wave, wave of minimum intensity is formed, and the respective fringe formed will be bright.(Destructive interference)

Application of Interference:

The phenomenon of interference can be observed in various phenomena such as illumination of colors in soap bubble, oil patch, peacock feathers etc.
When a thin transparent film is exposed to white light and seen in reflected light, different colors are seen in the film. These colors arise due to the interference of the light waves from the top and bottom surfaces of the film.
Consider a uniform transparent film having thickness t and refractive index μ. A ray of light AB incident at an angle i on the upper surface of the film is partly reflected along BC and partly reflected along BF at angle r. At point F the way BF is again partly reflected from the second surface along FD and partly emerge out along FK and so on. In this situation, the interference occurs between reflected waves BC and DE and also between transmitted waves FK and GL.



Analytical Treatment of Interference:
Let us consider two waves having the equation of displacement

$\psi_1 = a_1 sin\omega t$

and

$$\psi_2 = a_2 \sin(\omega t + \phi)$$

According to the principle of superposition we have,

$$\psi_r = \psi_1 + \psi_2$$

$$\psi_r = a_1 sin\omega t + a_2 sin(\omega t + \phi)$$

$$\psi_r = a_1 sin\omega t + a_2 sin\omega t cos\phi + a_2 sin\phi cos\omega t$$

Taking sin ωt common we have,

$$\psi_r = \sin\omega t\,(a_1 + a_2 cos\phi) + a_2 \sin\phi\ \cos\omega t$$

$$a_1 + a_2 cos\phi = A cos\theta$$

And

$$a_2 sin\phi = A sin\theta$$

Hence the equation now becomes

$$\psi_r = A \sin\omega t\, cos\theta + A sin\theta cos\omega t$$

Now this equation can be reduced as,

$$\psi_r = A \sin(\omega t + \theta)$$

Where

$$A = \sqrt{a_1^2 + a_2^2 + 2a_1 a_2 \cos\phi}$$

Where

$$\tan^{-1}\left(\frac{a_2 sin\phi}{a_1 + a_2 cos\phi}\right)$$

Condition for maximum intensity,

For maximum intensity, $\cos\phi = 1$

For $\phi = \pm 2n\pi$

Hence maximum intensity can be found to be

$$I_{max} = a_1^2 + a_2^2 + 2a_1 a_2$$

$$I_{max} = (a_1 + a_2)^2$$

If $a_1 = a_2 = a$

$$I_{max} = 4a^2$$

Condition for minimum intensity can be found by making the intensity to the minimum value,

$$\cos\phi = -1$$

This is possible when,

$$\phi = \pm(2n+1)\pi$$

Hence

$$I_{min} = a_1^2 + a_2^2 - 2a_1a_2$$

Or

$$I_{min} = (a_1 - a_2)^2$$

If $a_1 = a_2 = a$

Then

$$I_{min} = 0$$

For this maximum as well as minimum intensity, intensity vs phase plot can be drawn as shown in Fig.

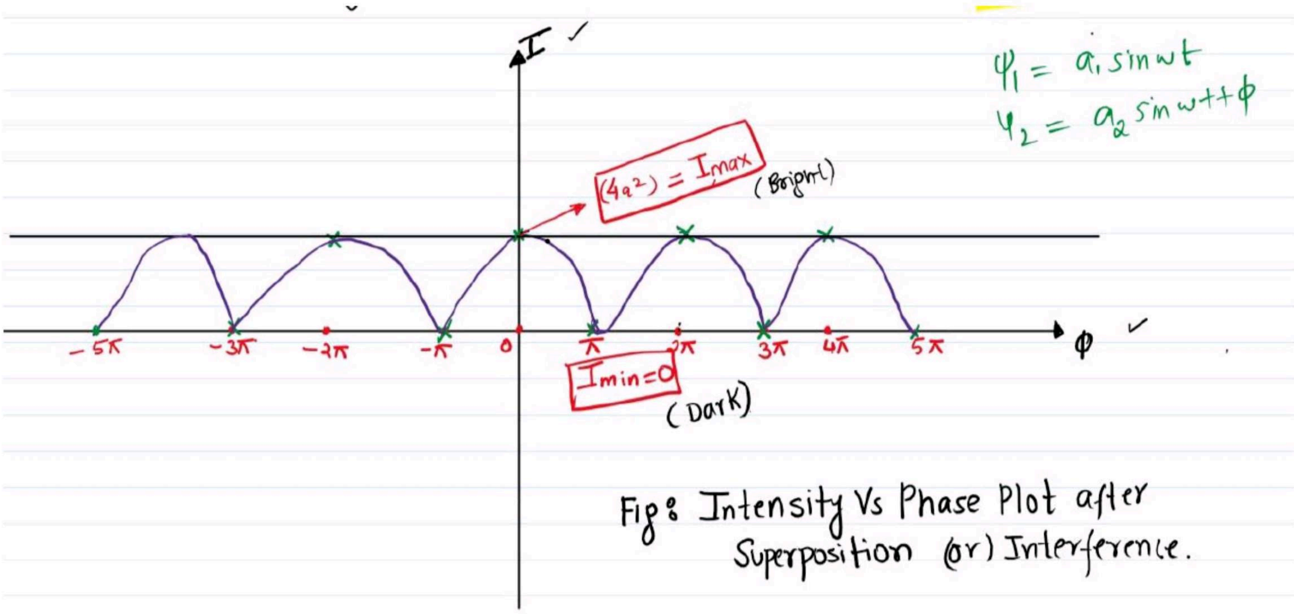


Fig: Intensity Vs Phase Plot after Superposition (or) Interference.

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LECTURE -2

Topic: - Michelson's Interferometer

Objectives:

1. With the application of He- Ne LASER, Alignment of Michelson's Interferometer is done in order to observe concentric circular fringes.
2. Using these circular fringes measurement of the wavelength of He-Ne LASER and Na lamp could be computed.
3. Equal inclination and equal thickness using Na lamp fringes study is performed.

Introduction:

In order to study the interference patterns with various light sources the devices so employed is called as interferometers. These could be categorized into two main classes:

1. On the basis of division of wave front.
2. On the basis of division of amplitude

Michelson interferometer: theory

While considering the Michelson interferometer the division of amplitude scheme is employed. Following principal measurements can be carried out with the aid of Michelson Interferometer:

- Spectral line width and fine structure.
- Lengths or displacements computed in terms of wavelengths of light.
- Transparent solid refractive indices could be computed.
- Differences in the velocity of light along 2 different directions.

With the aid of a beam splitter G1, the wave amplitude is divided by partial reflection, with the two resulting wave fronts maintaining the original width by having reduced amplitudes [1]. This beam splitter can be developed using plate of glass by making it partially reflective, and the splitting occurs part of the light is reflected off the surface and part of the light is transmitted through it.

In order to form the interference pattern the two beams that are obtained by amplitude division are sent in different directions against plane mirrors, then these are reflected back along the same respected path to the beam splitter. The set up for the Michelson interferometer consists of highly polished plates, A1 and A2 which act as the above mentioned mirror. G1 and G2 are the two parallel plates of glass - one is the beam splitter, and the other is a compensating plate, the function of which is discussed below. From mirror A1 the light that is reflected normally passes through G1 and reaches the eye. From A2 the light that is reflected, now passes through G2 for the second time, is reflected from the surface of G1 and into the eye.

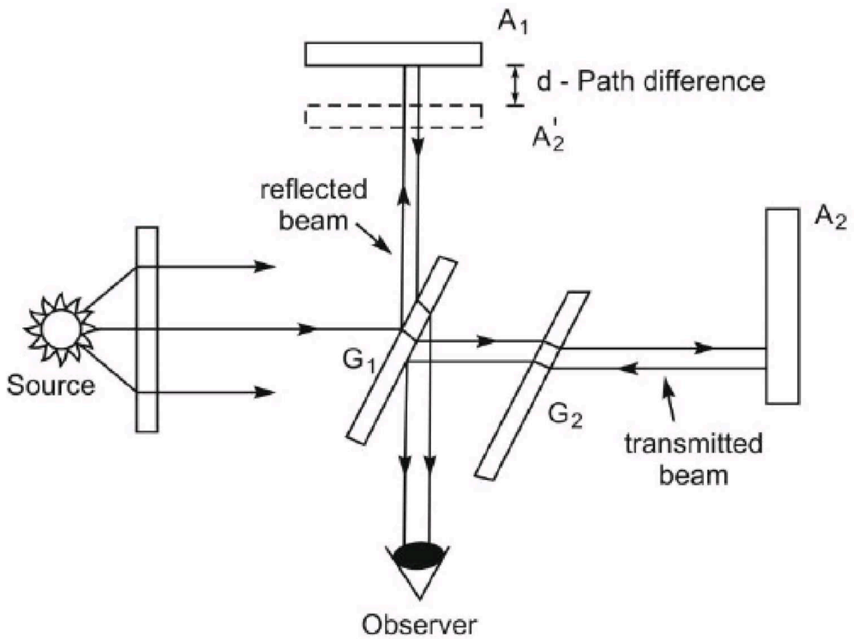


Figure 1: A schematic diagram of the Michelson interferometer.

The plate that is used in order to make the optical path equal is known as Compensating Plate and is represented as G2 in the given diagram.

[1]. However when we consider the monochromatic light this does not forms an essential part in order to produce effective, sharp and clear fringes in the monochromatic light, but when we consider production of fringes in white light this is crucial. With the aid of

micrometer the position of the mirror A1 can be adjusted which is mounted on a carriage. In order to obtain these fringes the mirrors are made exactly perpendicular with the aid of the calibration screws. The two primary requirements in order to fulfill interference fringes to appear are:

1. Use an *extended light source*. The crucial factor here is of illumination, it is evident that we may not be able to see the fringes on if the source is point, as there is not much space for you to see. Hence it becomes inevitable the use of extended source, generally by positioning a variable size aperture in front of an extended source and hence shrinking it's radius to minimum possible (thus effectively converting into point source). It could be seen that the field of view over which the fringes are observed shrinks right with it, hence it is inevitable to use the extended light source.

2. The light must be *monochromatic*, or nearly so. This is specifically important if the distances of A1 and A2 from G1 are significantly different. The relationship of the spacing f fringes for a given color of light to the wavelength of the light is that they are linearly proportional, hence it could be said that the fringes will coincide near to the region where the path difference is zero. The solid line here corresponds to the intensity of interference pattern of green light, and the dashed curve — to that of red light. It could be clearly observed that only around zero path difference colors remain relatively pure, only when we start to move farther away from that region will observe that colors will start to mix and start to mix hence becoming impure and unsaturated, already about 8-10 fringes away the colours mix back into white light, making fringes indistinguishable. Hence the region where fringes are visible is very narrow and hard to find with non-monochromatic light.

Sodium flame or a mercury arc is the some of the light sources suitable for the Michelson Interferometer. If instead small source bulb is utilized, a ground glass diffusing screen in front of the source shall do the job; through the plate G1 looking at the mirror A1, whole field of view filled with light is observed.

Circular Fringes

In order to carefully observe the circular fringes with the aid of monochromatic light, the mirror must be perfectly perpendicular to each other. The origin of the circular fringes is understood from Fig. 2.

The real mirror *A*2 has been replaced by its virtual image A'2 formed by the reflection in *G*1 : hence A'2 is parallel to *A1*.

Since light in the interferometer gets reflected many times, we can think of the extended source as being at *L*, where *L* is behind the observer as seen in Fig. 2; *L* forms 2 virtual images, *L*1 and *L*2, in mirrors *A*1 and *A2'*, respectively. The virtual sources in *L*1 and *L*2 are said to be in phase with each other (such sources are called *coherent sources*), in that the phases of corresponding points in the two are exactly the same at all times. If *d* is the separation of *A*1 and *A*2', the virtual sources are then separated by 2*d*, as can be seen in the diagram (Fig. 2).

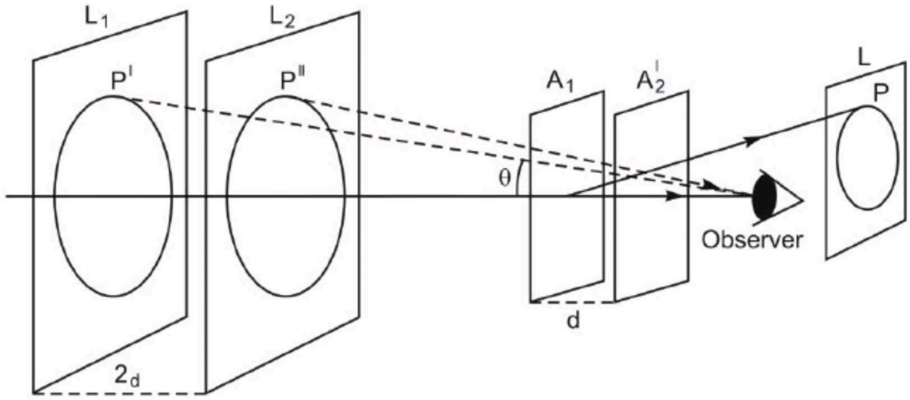


Figure 2: Virtual images from the two mirrors created by the light source and the beam splitter in the Michelson interferometer.

When *d* is exactly an integer number of half wavelengths, every ray that is reflected normal to the mirrors *A1* and *A'*2 will always be in phase. The path difference, 2*d*, must then be an integer number of wavelengths. Rays of light that are reflected at other angles will not, in general, be in phase. This means that the path difference between two incoming rays from points *P′* and *P″* will be $2d\cos\vartheta$, where ϑ is the angle between the viewing axis and the incoming ray. We can say that ϑ is the same for the two rays when *A1* and *A2*' are parallel, which implies that the rays themselves are parallel. Since the eye is focused to receive the parallel rays, it is more convenient to use a telescope lens, especially for looking at interference patterns with large values of *d*.

The parallel rays will interfere with each other, creating a fringe pattern of maxima and minima for which the following relation is satisfied:

$$2d\cos\vartheta = m\lambda \qquad (1)$$

Where *d* is the separation of *A1* and A'_2, *m* is the fringe order, λ is the wavelength of the source of light used, ϑ is as above (if the two are nearly collinear, we, of course, have

$\vartheta \approx 0$ — this is the case for the fringes in the very centre of the field of view).

Since, for a given *m*, λ, and *d* the angle ϑ is constant, the maxima and minima lie on a circular plane about the foot of the perpendicular axis stretching from the eye to the mirrors. As was mentioned before, the Michelson interferometer uses division by amplitude scheme: hence the resultant amplitudes of the waves, $\alpha 1$ and $\alpha 2$, are fractions of the original amplitude *A*, with respective phases $\alpha 1$ and $\alpha 2$. We can calculate the phase difference between the two beams based on the respective mirror separation. If the path difference is $2d\cos\vartheta$, then the phase difference δ for light of wavelength λ is simply

$$\delta = \frac{2\pi\,(2d\cos\theta)}{\lambda} \qquad (2)$$

Here the ratio of the path difference to the wavelength tells you what fraction of a wavelength have you passed, and multiplication by 2π makes it a fraction of the full period of a sinusoid, thus giving you exactly the sought phase difference.

By starting with *A1* a few centimeters beyond *A′2*, the fringe system will have the general appearance which is shown in Fig.3, where the rings of the system are very closely spaced. As the distance between *A1* and *A′2* decreases, the fringe pattern evolves, growing at first until the point of zero path difference is reached, and then shrinking again, as that point is passed.

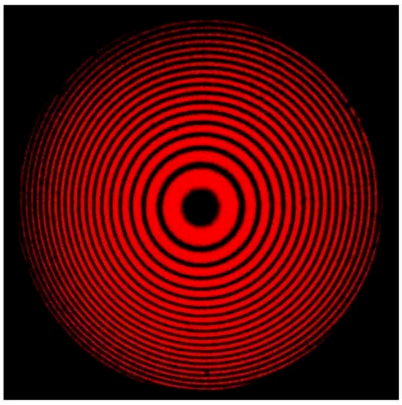


Figure 3: The circular fringe interference pattern produced by a Michelson interferometer.

This implies that a given ring, characterized by a given value of the fringe order *m*, must have a decreasing radius in order for (2) to remain true. The rings therefore shrink and vanish at the centre, where a ring will disappear each time $2d$ decreases by λ. This is because at the centre, $\cos\vartheta = 1$, and so we have the simplified version of equation (2),

$$2d = m\lambda \qquad (3)$$

From here we see that the fringe order changes by 1 precisely when $2d$ changes by λ, hence for a fringe to disappear we need to decrease $2d$ by λ, as claimed above.

Localized fringes

In case when the mirrors are not exactly parallel, fringes can still be observed in monochromatic light for path differences not much greater than a few millimeters. The space between the mirrors is wedge-shaped (Fig. 4): thus the two rays reaching the eye from the mirrors are no longer parallel and appear to diverge. Hence, the interference picture will be more like that of Fig. 5: the fringes are now semi-circles, with the centre lying outside the field of view — such fringes are often called *localized fringes*. The reason these fringes are almost straight is primarily because of the variation of the thickness of air in the wedge, as that is now the main reason for the variation of the path difference between the two beams across the field of view.

One would expect all fringes to be perfectly straight, parallel to the edge of the wedge: however, that is not the case, as the path difference still does vary somewhat with the angle ϑ, especially if d is large. Depending on the magnitude of d, we can observe different interference patterns: as we change the path difference the fringes become straighter, until we hit point of zero path difference. At that point, if we were looking at circular fringes, they would fill the whole field of view, become very large circles — that means that localized fringes would become parallel lines, as if there were small sections of the circumferences of very large circles. The association "large circular fringes — parallel localized fringes" will be important in the next section, when we use it to locate white light fringes.

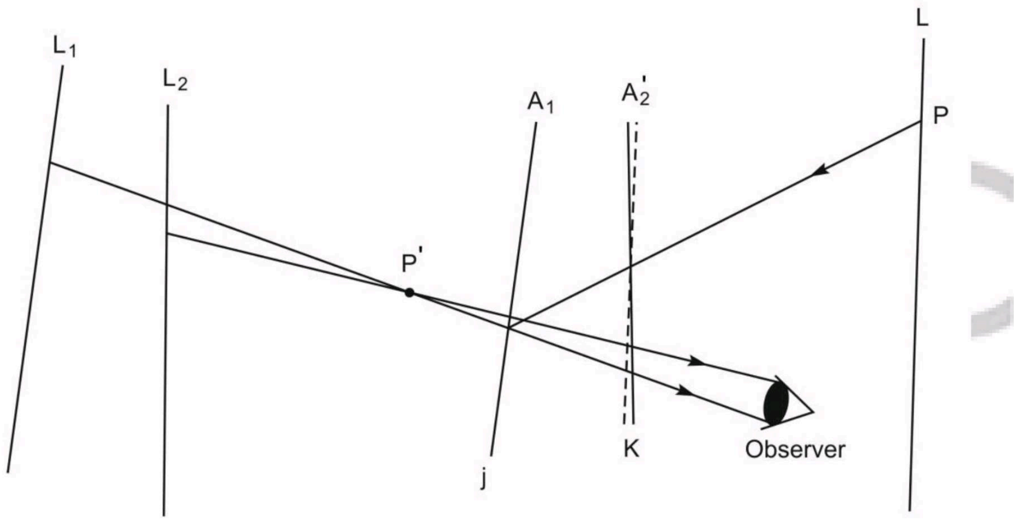


Figure 4: Formation of localized fringes with non-perpendicular mirrors — the air wedge is clearly seen.

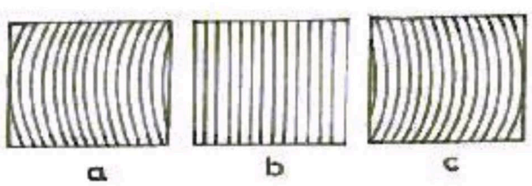


Figure 5: The localized fringe interference patterns produced by a Michelson interferometer: (a) and (c) are depictions of curved fringes, implying the mirror is far from the region of zero path difference, and (b) shows straight, parallel fringes — this must be at or very near the point of zero path difference.

White Light Fringes

If instead of using monochromatic light, we wish to study the fringes created by white light, no fringes will be seen at all except for when the path difference between *A1* and *A'*2 does not exceed a couple of wavelengths. They are extremely difficult to find, so *have patience* — they do exist.

This is well-demonstrated in Fig. 6: the dashed line corresponds to the intensity of the interference pattern of green light, while the solid line — to that of yellow light. As you can see from the diagram, the patterns only overlap over the narrow range of zero path difference between the two incoming beams: now if there are many different wavelengths involved in an interference process, as is the case for white light, one can conclude that anywhere too far away from the region of zero path difference the colours will mix back up into white light and no fringes will be visible.

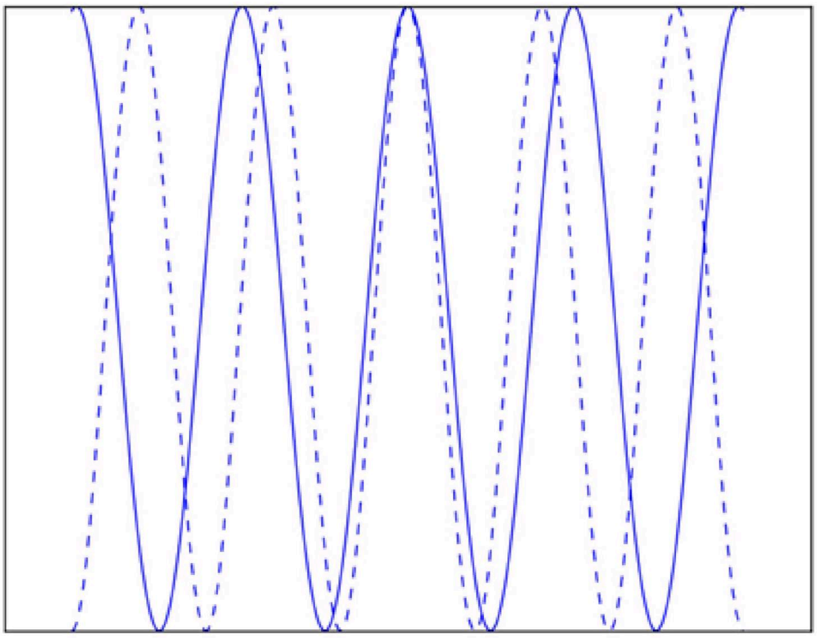


Figure 6: The intensity curves of interference of green light (dashed line) and yellow light (solid line). The dispersion of the interference patterns away from the region of zero path difference is readily observed.

With white light, there will be a central dark fringe, bordered on either side by 8 or 10 coloured fringes. Since the region over which the white light fringes are visible is so narrow, trying to search for it with white light alone is too time-consuming. Instead you can first approximate its location by using monochromatic light and finding the region of zero path difference. It will correspond to one of two regions: (a) if the mirrors are perfectly parallel and we are observing circular fringes, the region with the largest circular fringes is the region of zero path difference (b) if the mirrors are almost parallel and we are observing localized fringes, then the region with straight, parallel fringes will be the region of zero path difference. Once we have approximately found the right region, we switch back to white light, and move VERY slowly through the region: the bright fringes should come into view. These fringes will only occur over a very narrow range of path difference values, corresponding to about a 20 degree turn on the micrometer — hence the need to move slowly, otherwise you can miss them. We advise that, upon finding the fringes, you mark the approximate position of the micrometer, to simplify future search (as the position of white light fringes will be needed for other experiments).

Applications

1. The Michelson - Morley experiment is the best known application of Michelson Interferometer.
2. They are used for the detection of gravitational waves.
3. Michelson Interferometers are widely used in astronomical Interferometer.
4. The Michelson interferometer can be used to determine
   a) The wavelength of a given monochromatic source of light

b) The difference between the two neighboring wavelengths

c) Refractive index and thickness of a thin transparent film

d) Refractive index of liquids and gases

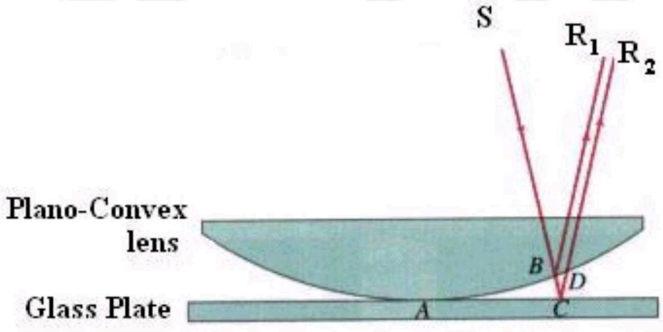
LECTURE -3

TOPIC: - NEWTON'S RINGS

NEWTON'S RINGS are the circular interference pattern first discovered by Newton.

FORMATION OF FRINGES

When a plano-convex lens with large radius of curvature is placed on a plane glass plate such that its curved surface faces the glass plate, a wedge air film (of gradually increasing thickness) is formed between the lens and the glass plate. The thickness of the air film is zero at the point of contact and gradually increases away from the point of contact.



If monochromatic light is allowed to fall normally on the lens from a source 'S', then two reflected rays R1 (reflected from upper surface of the film) and R2 (reflected from lower surface of the air film) interfere to produce circular interference pattern. This interference pattern has concentric alternate bright and dark rings around the point of contact.

Theory of Fringes:

The effective path difference between the two reflected rays R1 and R2 for a wedge shaped film from equation 2.18

$$\Delta = 2\mu t cos(r + \theta) + \frac{\lambda}{2}$$

If the light is incident normally on the lens, r = 0 and near to point of contact θ is small; therefore near point of contact, $(r + \theta) \to 0^o$

Therefore $\Delta = 2\mu t + \frac{\lambda}{2}$ 2.21

At point of contact t = 0 therefore the effective path difference Δ = λ/2 which is odd multiple of λ/2 **Therefore the Central fringe is dark.**

Condition of Maxima (Bright Fringe): The effective path difference $\Delta = \pm n\lambda$; substituting this in equation 2.21

$$2\mu t + \frac{\lambda}{2} = \pm n\lambda$$

$$2\mu t = \pm\frac{(2n-1)\lambda}{2}$$ 2.22

Condition for Minima (Dark Fringe): The effective path difference $\Delta = \pm\frac{(2n+1)\lambda}{2}$; substituting this in equation 2.21

$$2\mu t + \frac{\lambda}{2} = \pm\frac{(2n+1)\lambda}{2}$$

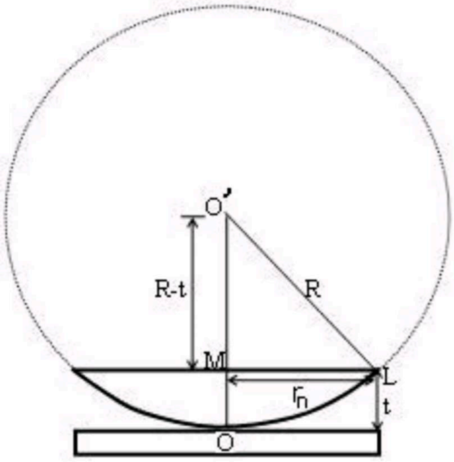
$$2\mu t = \pm n\lambda$$ 2.23

From 2.22 and 2.23 it is clear that for particular dark or bright fringe t should be constant.

Every fringe is the locus of points having equal thickness. Hence the fringes are circular in shape.

Diameter of Newton's Rings:

To calculate the diameter of fringes, assume a plano-convex lens is placed on a plane glass plate as shown in figure say R be the radius of curvature of lens. In ΔO'ML



$$(OL)^2 = (O'M)^2 + (ML)^2$$
$$R^2 = (R-t)^2 + r_n^2$$
$$R^2 = R^2 + t^2 - 2Rt + r_n^2$$
$$r_n^2 = 2Rt - t^2$$

As t << R; neglecting t^2 (small value)

$$r_n^2 = 2Rt$$

$$t = \frac{r_n^2}{2R} \quad ....2.24$$

Diameter of Dark Rings

Substitute t from eq. 2.24 to eq. 2.23

$$2\mu\frac{r_n^2}{2R} = n\lambda$$

$$r_n^2 = n\lambda R/\mu$$

As $D_n = 2r_n$

$$D_n^2 = 4n\lambda R/\mu$$

$$\boldsymbol{D_n = \sqrt{4n\lambda R/\mu}} \quad ....2.25a$$

The medium enclosed between the lens and glass plate is if air therefore, μ= 1. The diameter of n^{th} order dark fringe will be

$$\boldsymbol{D_n = \sqrt{4n\lambda R}}$$ 2.25b

$D_n \alpha \sqrt{n}$; n = 0, 1, 2, 3, 4....

The diameter of dark ring is proportional to square root of natural numbers

Diameter of Bright Rings

Substitute t from eq. 2.24 to eq. 2.22

$$2\mu.\frac{r_n^2}{2R} = \frac{(2n-1)\lambda}{2}$$

$$r_n^2 = \frac{(2n-1)\lambda R}{2\mu}$$

As $D_n = 2r_n$

$$D_n^2 = 2(2n-1)\lambda R/\mu$$

$$\boldsymbol{D_n = \sqrt{2(2n-1)\lambda R/\mu}}$$ 2.26a

The medium enclosed between the lens and glass plate is if air therefore, = 1. The diameter of n^{th} order bright fringe will be

$$\boldsymbol{D_n = \sqrt{2(2n-1)\lambda R}}$$ ....2.26b.

$D_n \alpha \sqrt{(2n-1)}$; n = 0, 1, 2, 3, 4....

The diameter of bright ring is proportional to square root of odd natural numbers

Spacing between Fringes

The Newton's rings are not equally spaced because the diameter of ring does not increase in the same proportion as the order of ring and rings get closer and closer as 'n' increases.

For example the diameter of dark ring is given by $D_n = \sqrt{4n\lambda R}$

Where n = 0, 1, 2, 3, 4....

$$D_5 - D_4 = (\sqrt{5} - \sqrt{4})\sqrt{\lambda R} = 0.236\sqrt{\lambda R}$$

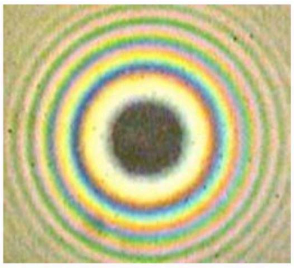
$$D_{25} - D_{24} = (\sqrt{25} - \sqrt{24})\sqrt{\lambda R} = 0.101\sqrt{\lambda R}$$

$$D_{15} - D_{14} = (\sqrt{15} - \sqrt{14})\sqrt{\lambda R} = 0.131\sqrt{\lambda R}$$

From above, we conclude that the fringe width reduces with increase in 'n'.

Newton's Ring with White Light

If the monochromatic source is replaced by the white light few coloured rings are seen around dark centre later illumination is seen in the field of view.

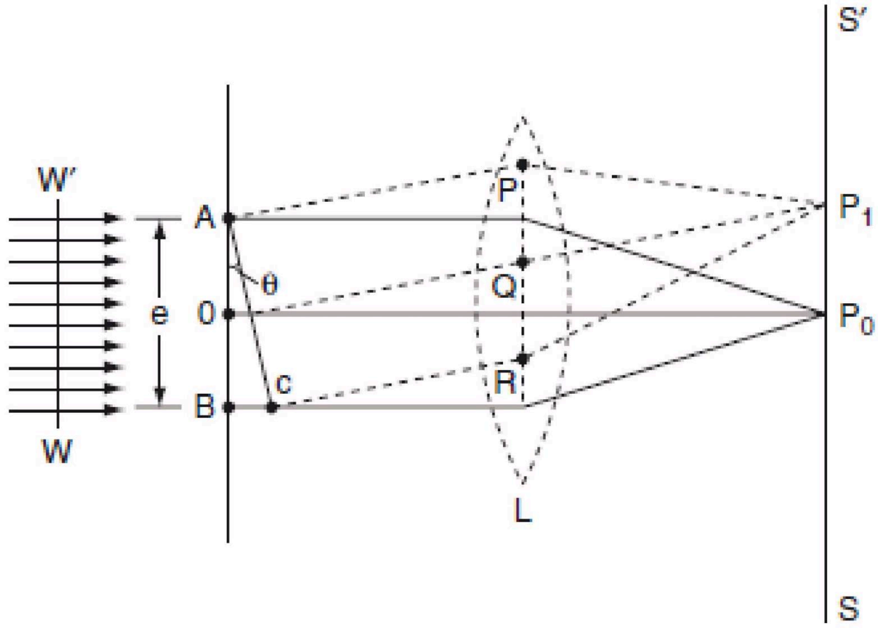


LECTURE -3

TOPIC: _-FRAUNHOFER DIFFRACTION FROM A SINGLE SLIT:-

Diffraction due to Single Slit

Description: The adjacent figure represents a narrow slit *AB* of width *'e'*. Let a plane wavefront of monochromatic light of wavelength 'λ' is incident on the slit. Let the diffracted light be focused by means of a convex lens on a screen. According to Huygen Fresnel, every point of the wavefront in the plane of the slit is a source of secondary wavelets. The secondary wavelets traveling normally to the slit *i.e., along* OP_o are brought to focus at P_o by the lens. Thus P_o is a bright central image. The secondary wavelets traveling at an angle 'θ' are focused at a point P_1 on the screen. The intensity at the point P_1 is either minimum or maximum and depends upon the path difference between the secondary waves originating from the corresponding points of the wavefront.



The intensity at the point P_1 is either minimum or maximum and depends upon the path difference between the secondary waves originating from the corresponding points of the wavefront.

Theory:

In order to find out the intensity at P_1, draw a perpendicular *AC* on *BR*.

The path difference between secondary wavelets from *A* and *B* in direction q is *BC i.e.* ,

$$\Delta = BC = AB sin\boldsymbol{\theta} = esin\boldsymbol{\theta}$$

So, the phase difference,

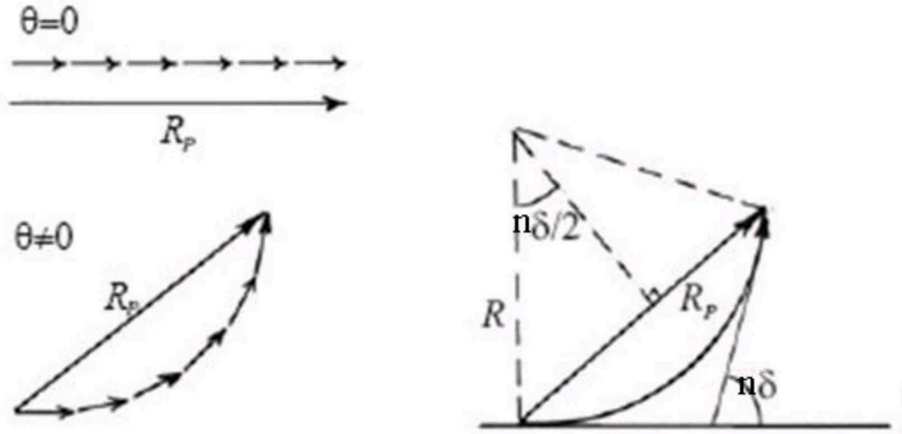
$$= \frac{2\pi}{\lambda} \times \Delta = \frac{2\pi}{\lambda}(esin\boldsymbol{\theta})$$

Let us consider that the width of the slit is divided into '*n*' equal parts and the amplitude of the wave from each part is *'a'*.

So, the phase difference between two consecutive points

$$\delta = \frac{1}{n} . \left\{ \frac{2\pi}{\lambda} (esin\boldsymbol{\theta}) \right\}$$ ....(2.32)

Then the *resultant amplitude R* is calculated by using the method of vector addition of amplitudes



The resultant amplitude of n number of waves having same amplitude 'a' and having common phase difference of 'δ' is

$$R = a \frac{sin(n\delta/2)}{sin(\delta/2)}$$ ....(2.33)

Substituting the value of δ in equation (2.33)

$$R = a \frac{sin\left(\frac{\pi}{\lambda}.esin\boldsymbol{\theta}\right)}{sin\left\{n\left(\frac{\pi}{\lambda}.esin\boldsymbol{\theta}\right)\right\}}$$ ....(2.34)

Substituting $\alpha = \frac{\pi}{\lambda}.esin\boldsymbol{\theta}$ in equation 2.34

$$R = a \frac{sin\alpha}{sin(\alpha/n)}$$

As α/n is small value; $sin\alpha/n \rightarrow \alpha/n$

$R = na\frac{sin\alpha}{\alpha}$ and na = A

Therefore

$$\boldsymbol{R = A\frac{sin\alpha}{\alpha}} \quad ....(2.35)$$

Therefore, the Intensity is given by

$$I^2 = R^2 = \boldsymbol{A^2\frac{sin^2\alpha}{\alpha^2}} \quad ....(2.36)$$

Case (i): ***Principal Maximum:***

Eqn (2.35) takes maximum value for

$\alpha = 0$

$\alpha = \frac{\pi}{\lambda}esin\theta = 0$

$\Rightarrow sin\theta = 0$ or $\theta = 0$

The condition

The condition $\theta = 0$ means that this maximum is formed by the secondary wavelets which travel normally to the slit along OP_o and focus at P_o. This maximum is known as *"Principal maximum"*.

Intensity of Principal maxima

$$R_{max} = Lim_{\alpha\rightarrow 0}\frac{Asin\alpha}{\alpha} = ALim_{\alpha\rightarrow 0}\frac{sin\alpha}{\alpha}$$

$$R_{max} = A.1 = A$$

Therefore

$$I^2_{max} = R^2_{max} = A^2$$

Case (ii): *Minimum Intensity positions:*

Eqn (2.34) takes minimum values for *sin*α*= 0.* The values of 'α' which satisfy $sin\alpha = 0$ are

$$\alpha = \pm\pi, \pm2\pi, \pm3\pi....... \pm n\pi$$

$$\frac{\pi}{\lambda}.esin\theta = \pm n\pi$$

$\boldsymbol{esin\theta = \pm n\lambda}$ where $n = 1, 2, 3, ...$...(2.37)

in the above eqn (2.37) *n = 0* is not applicable because corresponds to principal maximum. Therefore, the positions according to eqn (2.37) are on either side of the principal maximum.

Case (iii): *Secondary maximum:*

In addition to principal maximum at α = 0 , there are weak secondary maxima between minima positions. The positions of these weak secondary maxima can be obtained with the rule of finding maxima and minima of a given function in calculus. So, differentiating eqn (2.35) and equating to zero, we have

$$\because A^2 \neq 0; sin\alpha \neq 0$$

Because $sin\alpha = 0$ correspond to minima positions

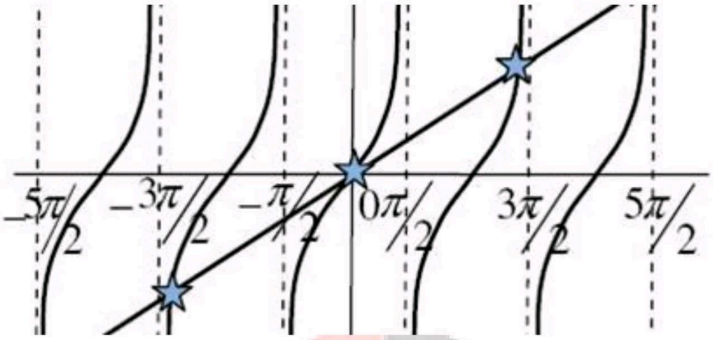
$$\therefore \alpha coc\alpha - sin\alpha = 0$$

$\Rightarrow \alpha = tan\alpha$..........(2.38)

The values of 'α' satisfying the eqn (2.38) are obtained graphically by plotting the curves $y = \alpha$ and $y = tan\alpha$

on the same graph. The points of intersection of the two curves gives the values of 'α' which satisfy eqn (2.38). The points of intersections are

$$\alpha = 0, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, \pm\frac{7\pi}{2} .... \pm \frac{(2n+1)\pi}{2}$$



But $\alpha = 0$, gives principal maximum, substituting the values of 'α' in eqn (2.36), we get

$$I_1 = A^2 \left[\frac{sin3\pi/2}{3\pi/2}\right]^2 = \frac{A^2}{22}$$

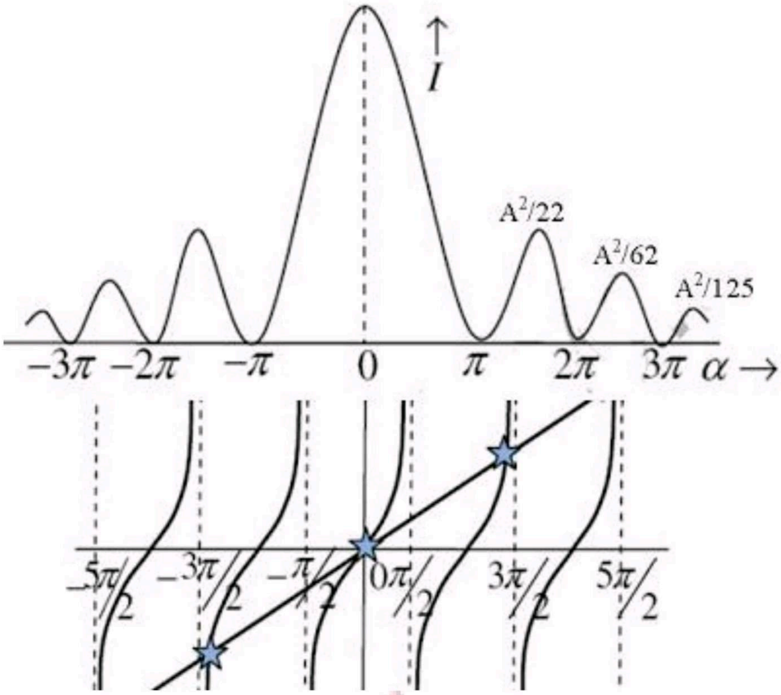
$$I_2 = A^2 \left[\frac{sin5\pi/2}{5\pi/2}\right]^2 = \frac{A^2}{62}$$

$$I_3 = A^2 \left[\frac{sin7\pi/2}{7\pi/2}\right]^2 = \frac{A^2}{125}$$

and so on. From the above expressions, I_{max}, I_1, I_2, I_3… it is evident that most of the incident light is concentrated at the principal maximum.

Intensity distribution graph:

A graph showing the variation of intensity with 'α' is as shown in the adjacent figure



LECTURE -4

TOPIC: _-DIFFRACTION GRATING: CONSTRUCTION, THEORY :-

DIFFRACTION OF LIGHT

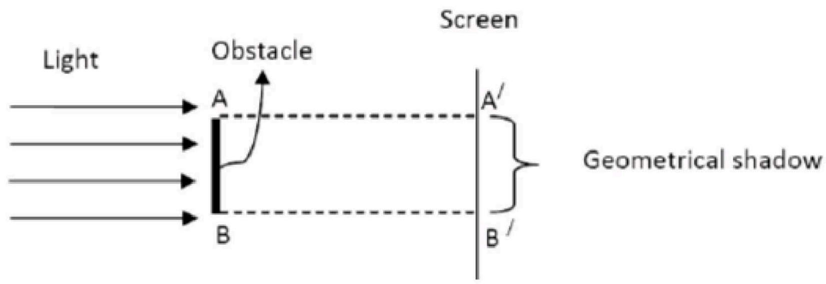
1. **Introduction**

The wave nature of light is first confirmed by the phenomenon of interference. Further it is confirmed by the phenomenon of diffraction. The word 'diffraction' is derived from the Latin word diffracts which means break to piece. When the light waves encounter an obstacle, they bend round the edges of the obstacle. The bending is predominant when the size of the obstacle is comparable with the wavelength of light. The bending of light waves around the edges of an obstacle is diffraction. It was first observed by **Francesco Gremaldy.**

2. Diffraction

When the light falls on the obstacle whose size is comparable with the wavelength of light then the light bends around the obstacle and enters in the geometrical shadow. This bending of light is called diffraction.

When the light is incident on an obstacle AB, their corresponding shadow is completely dark on the screen. Suppose the width of the slit is comparable to the wavelength of light, then the shadow consists of bright and dark fringes. These fringes are formed due to the superposition of bended waves around the corners of an obstacle. The amount of bending always depends on the size of the obstacle and wavelength of light used.



Distinction between interference and diffraction:

Interference

• Interference is the result of interaction of light coming from different wave fronts originating from the source.

• Interference fringes may or may not be of the same width.

• Regions of minimum intensity are perfectly dark.

• All bright bands are of same intensity

Diffraction

• Diffraction is the result of interaction of light coming from different parts of the same wave front.

• Diffraction fringes are not of the same width.

• Points of minimum intensity are not perfectly dark.

• All bright bands are not of the same intensity.

3. Types of diffraction

The diffraction phenomena are classified into two ways

I. Fresnel diffraction

II. Fraunhofer diffraction.

Fresnel diffraction:-

In this diffraction the source and screen are separated at finite distance. To study this diffraction lenses are not used because the source and screen separated at finite distance. This diffraction can be studied in the direction of propagation of light. In this diffraction the incidence wave front must be spherical of cylindrical.

Fraunhofer diffraction:-

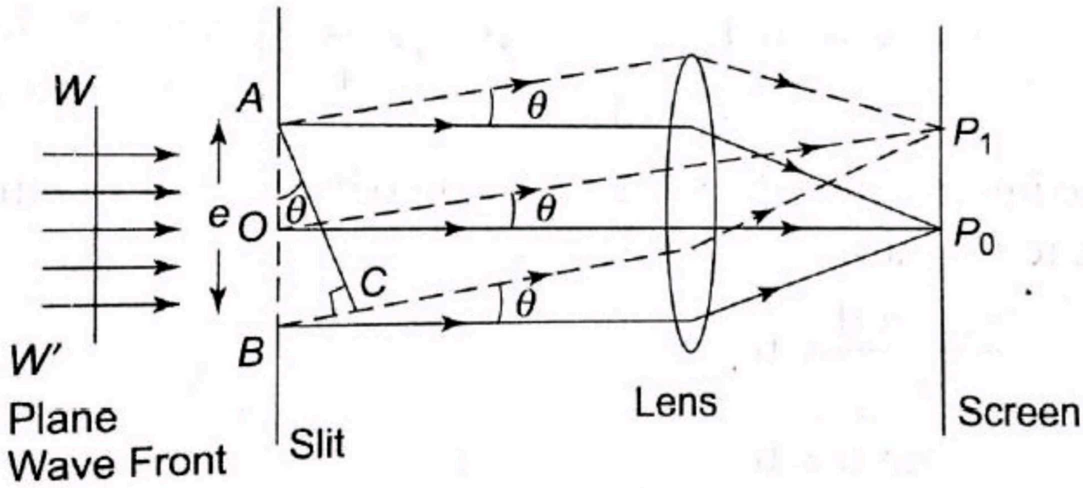
In this diffraction the source and screen are separated at infinite distance. To study this diffraction lenses are used because the source and screen separated at infinite distance. This diffraction can be studied in any direction. In this diffraction the incidence wave front must be plane.

4. Fraunhofer single slit diffraction:-

Let us consider a slit AB of width ‘e’.

Let a plane wave front ww' of monochromatic light of wavelength λ is incident on the slit AB.

According to Huygens principle, every point on the wave front is a source of secondary wavelets. The wavelets spread out to the right in all directions.



The secondary wavelets which are travelling normal to the slit are brought to focus at point P0 on the screen by using the lens.

These secondary wavelets have no path difference. Hence at point P0 the intensity is maxima and is known as central maximum. The secondary wavelets travelling at an angle θ with the normal are focused at point P1.

Intensity at point P1 depends up on the path difference between the wavelets A and B reaching to point P1. To find the path difference, a perpendicular AC is drawn to B from A.

The path difference between the wavelets from A and B in the direction of θ is

$path\ difference = BC = AB\sin\theta = e\sin\theta$

$phase\ difference = \frac{2\pi}{\lambda}(path\ difference) = \frac{2\pi(e\sin\theta)}{\lambda}$

Let the width of the slit is divided into ‘n’ equal parts and the amplitude of the wave front each part is ‘a’. Then the phase difference between any two successive waves from these parts would be

$\frac{1}{n}(phase\ difference) = \frac{1}{n}(2\pi e\sin\theta\lambda) = d$

Using the vector addition method, the resultant amplitude R is

$$R = \frac{a \sin \frac{nd}{2}}{\sin \frac{d}{2}}$$

$$R = A \frac{\sin \alpha}{\alpha} \qquad \because na = A \; and \; \alpha = \frac{\pi e \sin \theta}{\lambda}$$

Therefore resultant intensity $I = R^2 = A^2 \left(\frac{\sin \alpha}{\alpha}\right)^2$

Principal maximum:-

The resultant amplitude R can be written as

$$R = \frac{A}{\alpha}\left(\alpha - \frac{\alpha^3}{3!} + \frac{\alpha^5}{5!} - \frac{\alpha^7}{7!} + \cdots\right)$$
$$= \frac{A\alpha}{\alpha}\left(1 - \frac{\alpha^2}{3!} + \frac{\alpha^4}{5!} - \frac{\alpha^6}{7!} + \cdots\right)$$
$$= A\left(1 - \frac{\alpha^2}{3!} + \frac{\alpha^4}{5!} - \frac{\alpha^6}{7!} + \cdots\right)$$

In the above expression for α=0 values the resultant amplitude is maximum

$$R = A$$

Then $I_{max} = R^2 = A^2$

$$\alpha = \frac{(\pi e \sin\theta)}{\lambda} = 0$$

$$\sin\theta = 0$$

$$\theta = 0$$

For θ=0 and α=0 value the resultant intensity is maximum at P0 and is known as principal maximum.

Minimum intensity positions

I Will be minimum when $\sin\alpha = 0$

$$\alpha = \pm m\pi \qquad m = 1,2,3,4,5 \ldots \ldots$$
$$\alpha = \frac{\pi e \sin \theta}{\lambda} = \pm m\pi$$
$$\pi e \sin \theta = \pm m\lambda$$

So we obtain the minimum intensity positions on either side of the principal maxima for all $\alpha = \pm m\pi$ values

Secondary maximum

In between these minima secondary maxima positions are located. This can be obtained by differentiating the expression of I w.r.t α and equation to zero

$$\frac{dI}{d\alpha} = \frac{d}{d\alpha}\left(A^2\left[\frac{\sin\alpha}{\alpha}\right]^2\right) = 0$$

$$A^2\frac{2\sin\alpha}{\alpha}\frac{d}{d\alpha}\left(\frac{\sin\alpha}{\alpha}\right) = 0$$

$$A^2\frac{2\sin\alpha}{\alpha}\cdot\left[\frac{\alpha\cos\alpha - \sin\alpha}{\alpha^2}\right] = 0$$

In the above expression α can never equal to zero, so

Either $\sin\alpha = 0$ or

$\alpha\cos\alpha - \sin\alpha = 0$

$\sin\alpha = 0$ Gives the positions of minima

The condition for getting the secondary maxima is

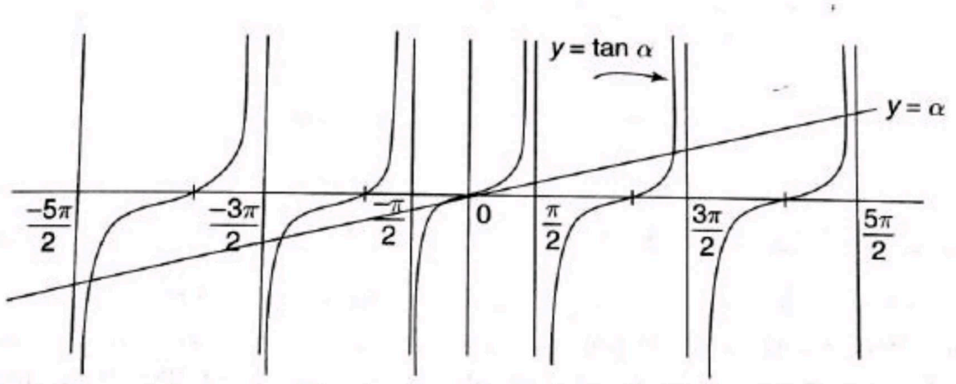
$\alpha\cos\alpha - \sin\alpha = 0$

$\alpha\cos\alpha = \sin\alpha$

$\alpha = \tan\alpha$

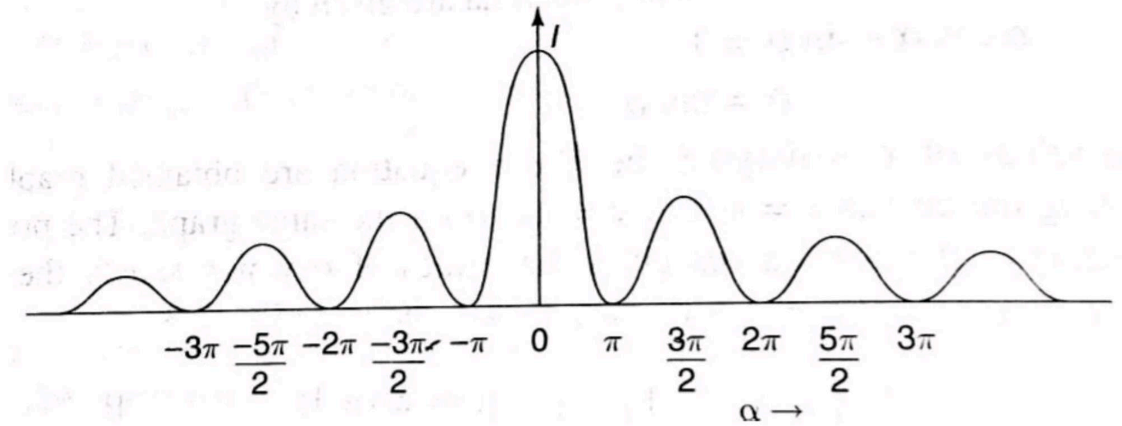
The values of α satisfying the above equation are obtained graphically by plotting the curves $Y=\alpha$ and

$Y=\tan\alpha$ on the same graph. The plots of $Y=\alpha$ and $Y=\tan\alpha$ is shown in figure.



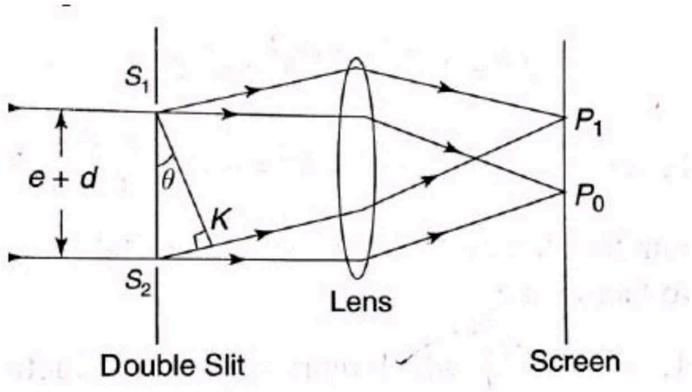
In the graph the two curves intersecting curves gives the values of satisfying of α satisfying the above equation. From the graph intersecting points are α = 0, ±$3\pi/2$, ±$5\pi/2$, ±$7\pi/2$…………

From the above concepts the intensity distribution curve verses α is shown in figure.



5. Fraunhofer double slit diffraction

- Let S1and S2 be the two slits equal width e and separated by a distance d.
- The distance between the two slits is (e + d).
- A monochromatic light of wavelength λ is incident on the two slits.
- The diffracted light from these slits is focused on the screen by using a lens.
- The diffraction of a two slits is a combination of diffraction and interference.
- When the plane wave front is incident on the two slits, the secondary wavelets from these slits travel in all directions.
- The wavelets travelling in the direction of incident light is focused at P0. The wavelets travelling at an angle θ with the incident light are focused at point P1.
- From Fraunhofer single slit experiment, the resultant amplitude is $R=A\sin\alpha\alpha$
- So the amplitude of each secondary wavelet travelling with an angle θ can be taken as $A\sin\alpha\alpha$. These two wavelets interference and meet at point P1 on the screen.



To find out the path difference between the two wavelets, let us draw a normal s1k to the wavelet S2. $Path\ difference$ = $S_2\ k$

From ΔS1 S2 k, $\sin\theta = \frac{(S2\ k)}{(S1S2)} = \frac{(S2\ k)}{(e+d)}$

S2 $k = (e+)\sin\theta$

$phase\ difference = \frac{(2\pi)}{(\lambda)}\ (path\ difference)$

$phase\ difference = \frac{(2\pi)}{(\lambda)}\ (e+)\ \sin\theta = \delta$

By using vector addition method, we can calculate the resultant amplitude at point P1 by taking the resultant amplitudes of the two slits S1and S2 as sides of the triangle. The third side gives resultant amplitude.

$$(AC)^2 = (AB)^2 + (BC)^2 + 2(AB)\ (BC)\cos\delta$$

$$R^2 = \left(A\frac{\sin\alpha}{\alpha}\right)^2 + \left(A\frac{\sin\alpha}{\alpha}\right)^2 + 2\left(A\frac{\sin\alpha}{\alpha}\right)\left(A\frac{\sin\alpha}{\alpha}\right)\cos\delta$$

$$R^2 = 4\left(A\frac{\sin\alpha}{\alpha}\right)^2 \cos^2{}_\beta$$

$$where\ \alpha = \frac{\pi e\sin\theta}{\lambda} and\ \beta = \frac{\pi(e+d)\sin\theta}{\lambda}$$

$$R^2 = 4\left(A\frac{\sin\alpha}{\alpha}\right)^2 \cos^2\beta$$

The resultant intensity $I = R^2 = 4\left(A\frac{\sin\alpha}{\alpha}\right)^2 \cos^2\beta$

From the above equation, it is clear that the resultant intensity is a product of two factors i.e.,

1. $(A\frac{\sin\alpha}{\alpha})^2$ Represents the diffraction pattern due to a single slit.

2. $\cos^2\beta$ Represents the interference pattern due to wavelets from double slit.

Diffraction effect

The diffraction pattern consists of central principal maxima for α=0 value.

The secondary maxima of decreasing intensity are present on either side of the central maxima for α=±3π/2, ±5π/2, ±7π/2 values.

Between the secondary maxima the minima values are present for α= ±π, ±2π, ±⋯ values.

In diffraction pattern the variation of I w.r.t α as shown in figure (a)

Interference effect

In interference pattern the variation of cos2β w.r.t β as shown in figure (b)

$\cos^2\beta$ Represent the interference pattern.

Interference maximum will occur for $\cos^2\beta$=1

$$\beta=\pm m\pi \text{ where } m=0,1,2,3,4\ldots\ldots\ldots$$

$$\beta=\pm\pi, \pm2\pi, \pm3\pi, \pm4\pi\ldots\ldots\ldots$$

$$\frac{\pi(e+d)\sin\theta}{\lambda} = \pm m\pi$$

$$(e+d)\sin\theta = m\lambda$$

Interference minima will occur for $\cos^2\beta$=0

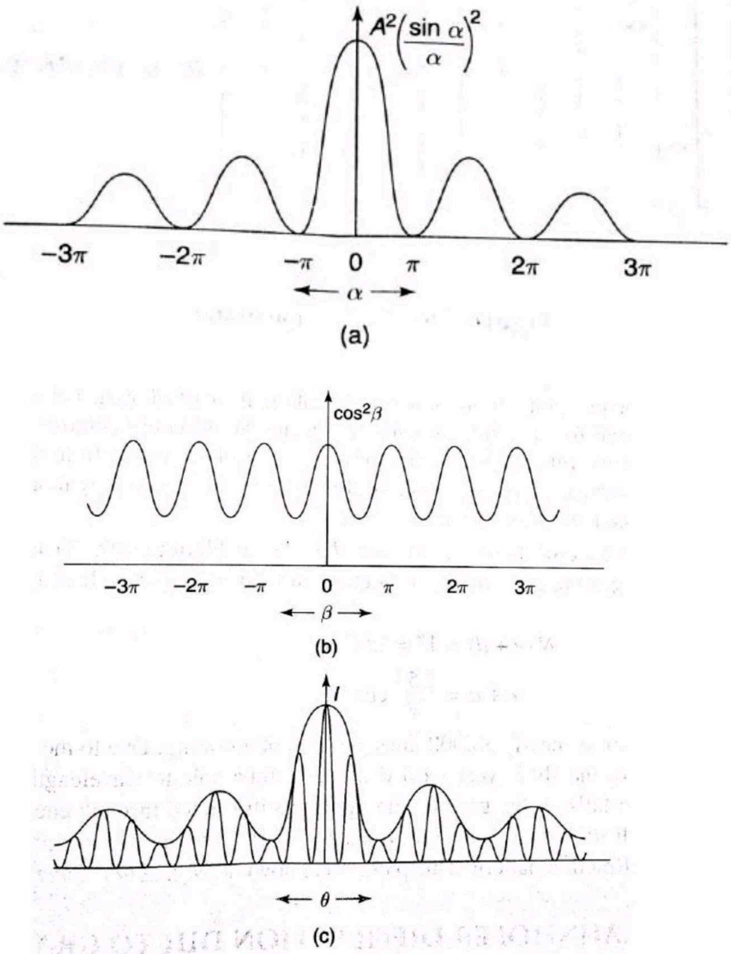
β= ± (2m+1) π/ 2

(e+)sinθ/ λ = ± (2m+1) π/ 2

(e+) sinθ = ±(2m+1)λ / 2

Intensity distribution:-

Figure (a), (b) and (c) represents the intensity variations of due to diffraction, interference and both effects respectively. From figure (c) it is clear that the resultant minima are not equal to zero.



Diffraction grating

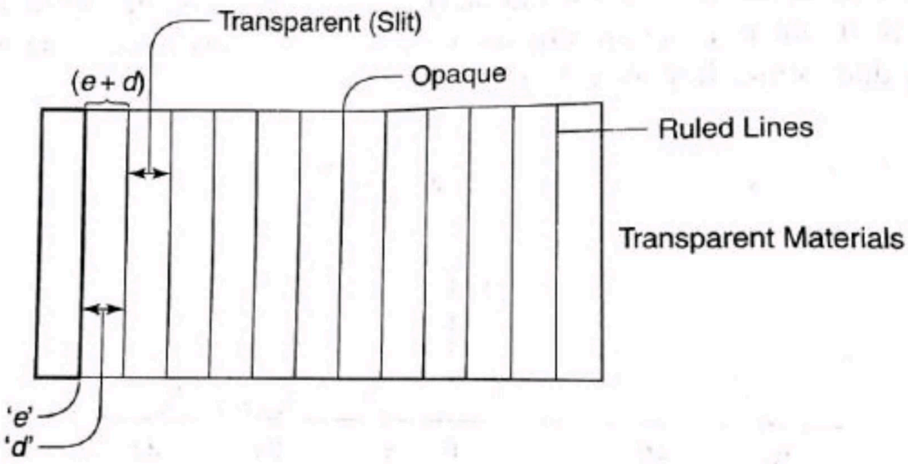
A set of large number of parallel slits of same width and separated by opaque spaces is known as diffraction grating. Fraunhofer used the first grating consisting of a large number of parallel wires placed side by side very closely at regular separation. Now the gratings are constructed by ruling the equidistance parallel lines on a transparent material such as glass with fine diamond point. The ruled lines are opaque to light while the space between the two lines is transparent to light and act as a slit.

Let 'e' be the width of line and'd' be the width of the slit. Then (e + d) is known as grating element. If N is the number of lines per inch on the grating then

$(e+d)$ =1 $inch$ = 2.54 cm

$(e+d) = \frac{2.54 \text{ cm}}{\text{N}}$

Commercial gratings are produced by taking the cost of actual grating on a transparent film like that of cellulose acetate. Solution of cellulose acetate is poured on a ruled surface and allowed to dry to form a thin film, detachable from the surface. This film of grating is kept between the two glass plates.



$$\text{or } 2d\theta_n = \frac{2\lambda}{N(b+d)\cos\theta_n} \qquad \text{(xix)}$$

This is the expression for angular width of the n^{th} order principal maxima, which shows that it depends on the total number of lines present on the grating and the wavelength of the light used in addition to the grating element.

2.11 APPLICATION OF DIFFRACTION GRATING

In the previous chapter we learnt how to determine the wavelength of monochromatic light using mechanism of interference. We can also find the wavelength of light by using the concept of diffraction with the help of diffraction grating.

2.11.1. Determination of Wavelength of Light

The theory of diffraction grating says that the principal maxima for a monochromatic light of wavelength λ is obtained as per the expression $(b + d) \sin \theta = n\lambda$. It is evident from this expression that if we are able to accurately measure the angle of diffraction θ and the grating element $(b + d)$, which is equal to the reciprocal of the number of lines per cm, we can determine the wavelength of light. A typical arrangement of using diffraction grating for the measurement of light is depicted in Fig. 2.20.

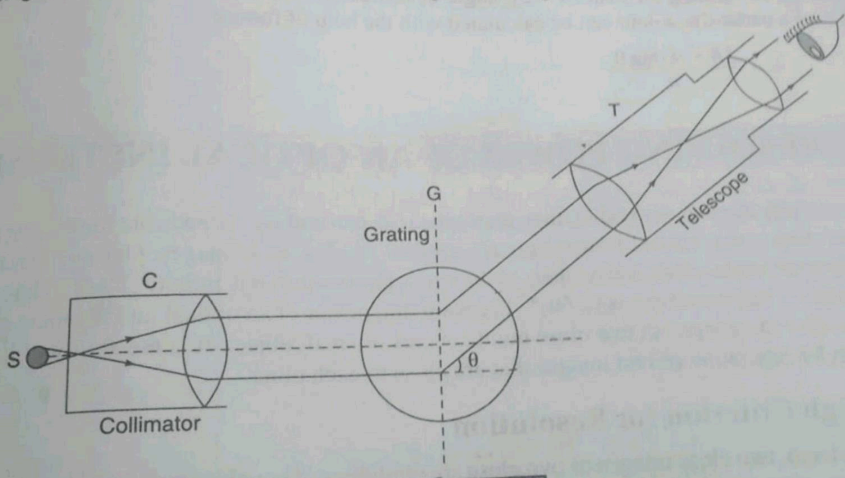


Fig. 2.20

In place of a monochromatic light, if we use the ordinary light (which has different wavelengths), then the beam gets dispersed (diffracted) by the grating. Under this situation the spectrum of constituent wavelengths is obtained, as shown in Fig. 2.21. This is due to the different angle θ for different wavelength λ for satisfying the condition of principal maxima.

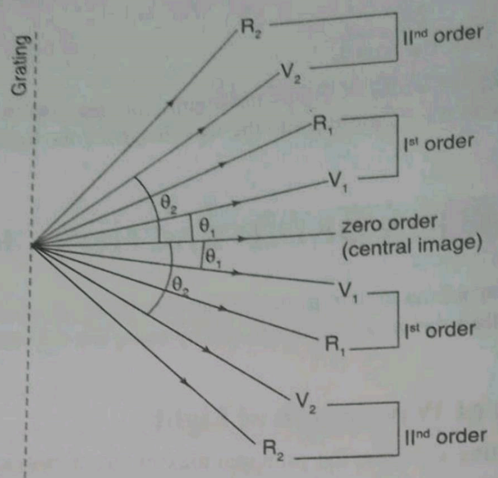


Fig. 2.21

Thus by knowing the grating element $(b + d)$, angle of diffraction θ and the order n of the diffraction, the wavelength λ of a particular colour can be calculated with the help of formula

$$\lambda = \frac{(b+d)\sin\theta}{n}$$